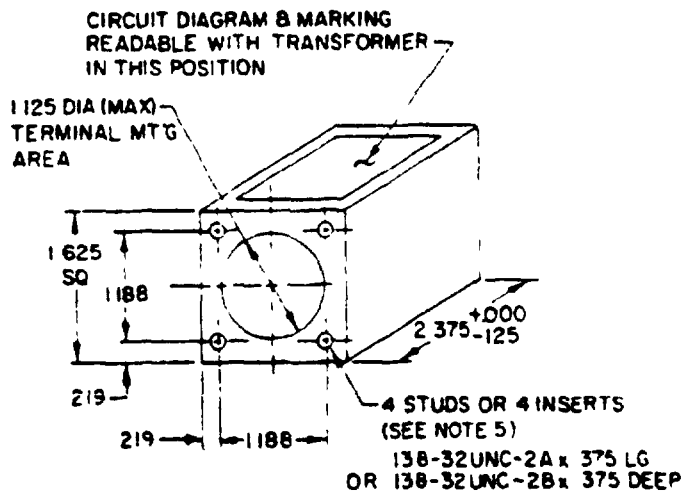
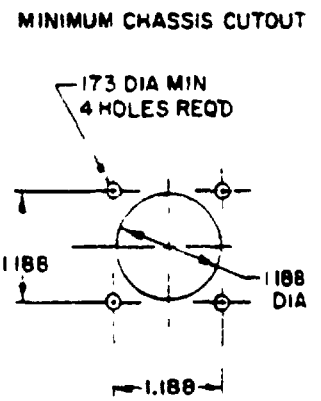
MILITARY SPECIFICATION SHEET

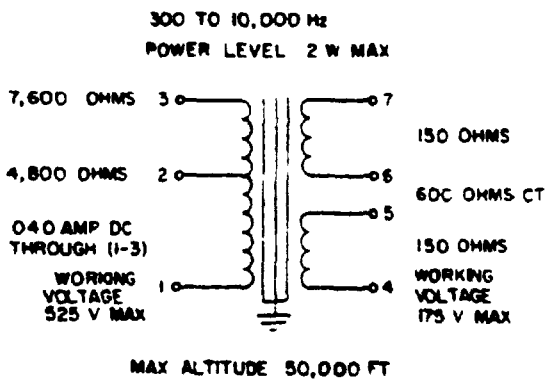
Ⓐ TRANSFORMER, AUDIO FREQUENCY, TF4R21AJ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.







CIRCUIT DIAGRAM and Marking

INCHES	MM
.125	3.18
.138	3.51
.219	5.56
.375	9.53
1.125	28.58
1.188	30.18
1.625	41.28
2.375	60.33

NOTES:
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm
3. Unless otherwise specified, tolerance on overall case dimensions is +.000 (.00 mm) -.063 (1.60 mm).
4. Tolerance on mounting dimensions is +.016 (.41 mm).
5. Inserts not recommended for new design
6. Terminal arrangement optional within mounting area indicated.
7. When numbers in parentheses, e.g. (1-3) are used, they indicate the winding and the extreme terminals of the winding. When the extreme terminals of two windings are used, e.g. (4-7) the windings are connected in series, i.e., terminals 5 and 6 are connected.

Ⓐ denotes change

FSC 5950

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REQUIREMENTS:

Electrical rating

Primary power level - 2 W, maximum, .1 W, minimum.

Working voltage (1-3) - 525 V, maximum.
(4-5) - 175 V, maximum.
(6-7) - 175 V, maximum.

Primary current - .040 ampere dc through terminals (1-3).

Primary impedance (1-3) - 7,600 ohms.
(1-2) - 4,800 ohms.

Design, configuration, and physical dimensions: See figure.

Material: Metal encased, steel.

Weight .72 pound, maximum.

Duty cycle: Continuous.

Altitude: 50,000 feet, maximum.

Life expectancy 10,000 hours, minimum.

Temperature range: -55°C to 105°C.

Terminals: Solder terminals, No. 18 AWG.

Terminal height: .375 (9.53 mm) $^{+.000}_{-.188}$ (4.78 mm).

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Operating temperature: 105°C, maximum.

Dielectric withstanding voltage: Method 301 of MIL-STD-202; test voltage at sea level, (1-3) 1,500 Vrms, (4-5) 500 Vrms, (6-7) 500 Vrms. At reduced barometric pressure, (1-3) 660 Vrms, (4-5) 300 Vrms, (6-7) 300 Vrms.

Electrical characteristics:

Harmonic distortion - Not more than 1 percent with 124 V, 300 Hz across (1-3) and .040 ampere dc through (1-3).

No load (center-tap voltage unbalance only) - With 124 V, 1,000 Hz across (1-3) and .040 ampere dc applied to (1-3):
Unbalance (4-5) and (6-7) - 1 percent, maximum.

Secondary load impedance (4-5) - 150 ohms.
(6-7) - 150 ohms ±10 percent.
(4-7) - 600 ohms ct.

Primary impedance - With approximately 5 V, 1,000 Hz and .040 ampere dc applied to (1-3) and 600 ohms ct across (4-7), the primary impedance shall be 7,600 ohms ±10 percent across (1-3).

Frequency response:
Range - 300 to 10,000 Hz at rated power, flat within ±2 dB; reference frequency = 1,000 Hz; test voltage 5 Vrms (1-3).

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Insertion loss - 1.5 dB maximum, with 124 Ω, input at 1,000 Hz across (1-3).

Resonance - Second resonant frequency = 20,000 Hz minimum.

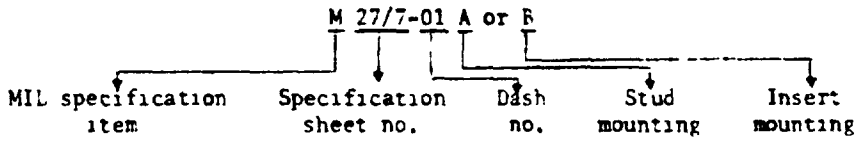
Polarity - Additive, with terminals 3 and 4, and 5 and 6 connected.

Temperature rise: 20°C maximum, above an ambient of 85°C, with rated power at 300 Hz.

Vibration, high frequency: Method 204, test condition B.

Shock (specified pulse): Method 213, test condition I.

Part number example:



(For cross reference of MS part number to military part number, see table I.

TABLE I. Cross reference.

Mounting	MS90004B		M27/7	
	Dash no.	Type designation	Dash no.	Type designation
Studs	-A	TF4R\13AJ001	-01A	TF4R21AJ
Inserts	-B	TF4R\13AJ001	-01B	TF4R21AJ

(A) NOTE: Qualification approval to MIL-T-27/7 shall be sufficient to grant qualification approval to MIL-T-27/3 through MIL-T-27/6 and MIL-T-2./8.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI
Navy - EC
Air Force - 11, 17, 99
DSA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5950-0533-4)

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